

Development and internal validation of a preliminary preoperative prediction model for postoperative transoral endoscopic thyroidectomy vestibular approach hypocalcemia

(Title: 13 words — within the 20-word limit)

Authors

[Author 1 Full Name]¹, [Author 2 Full Name]¹, [Author 3 / Engineering Team Member]², [Senior Author / Supervisor]^{1*}

* Corresponding author

Affiliations

¹ [Department, Faculty/Institution, City, Country]

² [Department of Computer Engineering, Institution, City, Country]

Corresponding author contact

Name: [PENDING]

Email: [PENDING]

Address: [PENDING]

Phone: [PENDING]

○ Title page — per ICTMHS 2026 guideline, manuscript text file must start with a title page showing author affiliations and contact information, with the corresponding author identified by an asterisk. Please replace all placeholders before submission.

Abstract

(230 words — within the 250-word limit; structured per IMRAD; no references included, per guideline)

Introduction: Postoperative hypocalcemia remains a common complication following transoral endoscopic thyroidectomy vestibular approach (TOETVA). A key limitation of existing prediction models is their dependence on intraoperative or early postoperative parathyroid hormone (PTH) levels. Thus, we aimed to develop and internally validate a preliminary preoperative prediction model for post-TOETVA hypocalcemia.

Methods: We conducted a retrospective cohort study of 42 patients who underwent TOETVA. Hypocalcemia occurred in 7 patients (16.7%). Three preoperative predictors—PTH, calcium, and vitamin D—were evaluated. Given the small sample size and low event count (events-per-variable ≈ 2.3), three logistic regression-based models were developed and compared: standard logistic regression, Firth logistic regression, and LASSO logistic regression. Internal validation was performed using bootstrap resampling (1,000 resamples) to derive optimism-corrected area under the receiver operating characteristic curve.

Results: [PENDING] The optimism-corrected AUC was [X.XX, 95% CI X.XX–X.XX] for standard logistic regression, [X.XX] for Firth logistic regression, and [X.XX] for LASSO logistic regression. The [MODEL NAME] model showed the most stable performance.

Conclusion: This preliminary cohort study suggests a preoperative-only model may offer [promising/limited] discrimination for post-TOETVA hypocalcemia. However, these findings should be considered hypothesis-generating and validation requires prospective, multicenter validation in larger patient populations.

Keywords

Transoral endoscopic thyroidectomy vestibular approach; hypocalcemia; prediction model; Firth regression; LASSO; internal validation

(6 keywords — within the 6-keyword limit; "small sample" removed to comply)

Introduction

Transoral endoscopic thyroidectomy vestibular approach (TOETVA) has emerged as an alternative to conventional thyroidectomy, offering comparable oncologic safety with improved cosmetic outcomes. As with conventional thyroid surgery, postoperative hypocalcemia remains one of the most common complications following TOETVA, resulting from inadvertent injury, devascularization, or removal of the parathyroid glands during dissection through the narrow transoral working space. Given the technically demanding nature of the approach and its relatively recent adoption, robust preoperative risk stratification tools specific to TOETVA are lacking.

Prior work in conventional thyroidectomy has consistently identified perioperative parathyroid hormone (PTH) levels as the strongest predictors of postoperative hypocalcemia, outperforming clinical and surgical variables in both regression-based and machine learning models [1,2]. However, these models have generally relied on intraoperative or early postoperative PTH measurements, which are not available at the time of preoperative counseling and surgical planning. A prediction model based on preoperative variables including PTH, calcium, and vitamin D would allow risk stratification before surgery, potentially informing patient counseling, operative planning, and the threshold for prophylactic calcium/vitamin D supplementation.

A central methodological challenge in developing such a model is the limited sample size inherent to TOETVA cohorts, given the procedure's relatively recent introduction and lower-case volume compared with conventional thyroidectomy. In our study, the events-per-variable (EPV) ratio for a three-predictor model is approximately 2.3, well below the commonly cited threshold of 10 EPV historically recommended for stable logistic regression estimation. Under these conditions, standard maximum-likelihood logistic regression is prone to complete or quasi-complete separation and finite-sample bias, particularly with rare outcomes. Firth's penalized-likelihood logistic regression was developed specifically to address this bias in small-sample and rare-event settings, while least

absolute shrinkage and selection operator (LASSO) regression offers an alternative regularization approach that shrinks coefficients toward zero, reducing overfitting and improving out-of-sample generalizability at the cost of introducing bias [3,4,5,6].

Given this, **the present study did not aim to establish a definitive clinically prediction model**, but rather to conduct a preliminary, hypothesis-generating comparison of three logistic regression-based approaches including standard logistic regression, Firth logistic regression, and LASSO logistic regression to identify which modeling strategy offers the most stable and generalizable performance for predicting post-TOETVA hypocalcemia using preoperative biochemical variables alone **to prepare for future prospective and adequately powered validation**.

The objectives of this study were: (1) to describe the association between preoperative PTH, calcium, and vitamin D levels and the occurrence of post-TOETVA hypocalcemia in a retrospective cohort; (2) to develop and internally validate three candidate prediction models using bootstrap resampling; and (3) to compare the discriminative performance and stability of the three models and identify the most suitable approach for this small-sample, preoperative prediction context.

Methods

Study Design and Population

This was a retrospective cohort study conducted at **[INSTITUTION]**, including consecutive patients who underwent TOETVA between **[START DATE]** and **[END DATE]**. Reporting follows the TRIPOD statement for prediction model development, adapted for a preliminary internal-validation-only study without external validation.

All patients who underwent TOETVA at the study institution during the study period were considered for inclusion.

Inclusion criteria: **[PENDING]**

Exclusion criteria: **[PENDING — e.g., preoperative hypocalcemia, missing preoperative laboratory data, revision surgery]**

The final cohort comprised 42 patients.

Ethics Declaration

This study was approved by **[institutional/licensing ethics committee name and protocol number — PENDING]**. All procedures were performed in accordance with the relevant institutional guidelines and regulations and the 1964 Declaration of Helsinki and its later amendments. **All patient data were de-identified prior to analysis. No study participant names or other personally identifiable information are included in the text, figures, or tables of this manuscript.**

Outcome Definition

The primary outcome was postoperative hypocalcemia, defined as **[PENDING: exact threshold and timing]**. Hypocalcemia occurred in 7 patients (16.7%); the remaining 35 patients did not develop hypocalcemia.

Predictor Variables

Three preoperative biochemical predictors were selected a priori based on biological plausibility and prior literature on calcium homeostasis: preoperative intact parathyroid hormone (PTH), preoperative serum calcium, and preoperative vitamin D (25-hydroxyvitamin D). Age and sex were additionally recorded as descriptive/optional covariates but were not included as primary model predictors, given **the limited sample size** and the a priori focus on biochemical predictors.

Statistical Analysis

Data completeness and the pattern of missing data were assessed for all candidate predictors. The distribution of each continuous variable was examined visually and assessed for normality using the Shapiro-Wilk test. Patients were stratified into hypocalcemia group (n=7) and no hypocalcemia group (n=35), and continuous variables were summarized as median (interquartile range) or mean $\pm$ standard deviation according to their distribution while categorical variables were summarized as frequency (percentage). Between-group differences were assessed using the Mann–Whitney U test for continuous variables and Fisher's exact test for categorical variables.

Three logistic regression-based models including standard logistic regression, Firth logistic regression, LASSO logistic regression were developed with all three preoperative predictors. All continuous predictors were standardized prior to model fitting.

Internal validation was performed using bootstrap sample with 1000 resamples from the original 42-patient cohort, following the approach described by Harrell et al. and Steyerberg et al. for optimism correction in small-sample models [7,8]. For each resample, the models were refit and evaluated on both the bootstrap sample (apparent performance) and the original sample (test performance). The average difference between these estimates across all iterations served as the optimism estimate, which was subtracted from the apparent performance to yield optimism-corrected area under the receiver operating characteristic (ROC) curve (AUC) values with 95% confidence intervals (CI).

For each model, the following performance metrics were reported: AUC with 95% CI (primary metric); sensitivity and specificity at a pre-specified/optimal probability threshold [PENDING: e.g., Youden's index]; and ROC curves for visual comparison. As exploratory/secondary metrics, calibration plots and the Brier score were additionally reported where feasible, acknowledging the limited precision of calibration assessment given the small number of outcome events. The three models were compared based on their bootstrap-derived, optimism-corrected AUC and the width of the corresponding 95% CI as a measure of estimate stability. The preferred model was selected based on (a) highest optimism-corrected AUC, (b) narrowest/most stable CI, and (c) simplicity of the final model, with simplicity prioritized in the event of comparable performance — consistent with methodological guidance favoring parsimonious models in small-sample prediction research.

All statistical analyses were performed using [PENDING: R version X.X (packages: logistf, glmnet, pROC/rms) / Python version X.X]. A two-sided p value <0.05 was considered statistically significant for group comparisons; given the exploratory nature of this step, no correction for multiple comparisons was applied.

Results

[PENDING — awaiting completed analysis]

Study population

[PENDING — flow of patient inclusion, baseline data completeness]

Descriptive and group comparison

Table 1 summarizes preoperative characteristics by hypocalcemia status. [PENDING — narrative summary of Mann-Whitney/Fisher's exact results]

Variable	Hypocalcemia (n=7)	No hypocalcemia (n=35)	P value
PTH (pg/mL)	[PENDING]	[PENDING]	[PENDING]
Calcium (mmol/L)	[PENDING]	[PENDING]	[PENDING]
Vitamin D (ng/mL)	[PENDING]	[PENDING]	[PENDING]
Age (years)	[PENDING]	[PENDING]	[PENDING]
Sex, female n (%)	[PENDING]	[PENDING]	[PENDING]

Internal validation performance

Table 2 summarizes the optimism-corrected AUC, sensitivity, and specificity for each of the three models. [PENDING — narrative summary]

Model	Optimism-corrected AUC (95% CI)	Sensitivity / Specificity (%)
Standard logistic regression	PENDING	PENDING
Firth logistic regression	PENDING	PENDING
LASSO logistic regression	PENDING	PENDING

Model comparison and selected model

[PENDING — statement of which model was chosen and rationale, referring to Figure 1 (ROC curves) and Figure 2 (calibration plot, if feasible)]

Discussion

[PENDING — to be completed once results are available; per guideline, Discussion should be written as continuous text without subheadings]

[PENDING] This preliminary study compared three logistic regression-based approaches for predicting post-TOETVA hypocalcemia using preoperative PTH, calcium, and vitamin D in a small retrospective cohort. [Summary of which model performed best/most stably and the direction of association for each predictor to be inserted here.]

[PENDING] These findings should be interpreted in the context of the broader methodological literature on small-sample clinical prediction modeling. Given an events-per-variable ratio of approximately 2.3 in the present cohort, standard logistic regression was expected to be susceptible to finite-sample bias and unstable coefficient estimates, motivating the inclusion of Firth and LASSO logistic regression as bias-corrected and regularized alternatives, respectively [3,4,5,6]. [Discussion of whether the observed results aligned with this theoretical expectation to be inserted here.]

[PENDING] Compared with prior post-thyroidectomy hypocalcemia prediction models, which have generally relied on intraoperative or early postoperative PTH measurements as the dominant predictor [1,2], the present model was deliberately restricted to preoperative variables in order to address an earlier point in the clinical decision-making pathway—namely, preoperative counseling and surgical planning—rather than postoperative management. [Discussion of clinical implications specific to TOETVA to be inserted here.]

This study has several important limitations. The sample size was very small, with only 7 hypocalcemia events, limiting statistical power and the precision of all reported performance estimates; the wide confidence intervals anticipated around the AUC estimates should therefore be interpreted with caution. The retrospective, single-center design may limit generalizability, and only internal validation was performed, without an external validation cohort. Given these limitations, the title and framing of this work as “preliminary” is intentional, and the findings should be regarded as hypothesis-generating rather than confirmatory. Future work should prioritize prospective, multicenter data collection to achieve an adequate events-per-variable ratio, followed by external validation of the selected model, and may consider the inclusion of additional preoperative variables to further improve discrimination.

References

(Vancouver style, numbered sequentially in order of first citation in text — please verify exact volume/page details against the original sources before submission)

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Author Contributions

[Author 1]: conceptualization, methodology, writing – original draft.

[Author 2]: data curation, formal analysis, writing – original draft.

[Author 3 / Engineering]: software, formal analysis, writing – review & editing.

[Senior author]: supervision, writing – review & editing.

All authors have approved the submitted version and agree to be personally accountable for their own contributions.

Additional Information

Competing Interests

The author(s) declare no competing interests. [PENDING — amend if any author has a relevant financial or non-financial interest to declare]

Funding

[PENDING — state funding source, or: "This research received no specific grant from any funding agency."]

Figure Legends

(Figure legends must be placed after References, per guideline; maximum 350 words per legend)

Figure 1. [PENDING] Receiver operating characteristic (ROC) curves comparing the discriminative performance of standard logistic regression, Firth logistic regression, and LASSO logistic regression for predicting post-TOETVA hypocalcemia, based on optimism-corrected bootstrap internal validation (1,000 resamples).

Figure 2. [PENDING] Calibration plot(s) comparing predicted versus observed probability of post-TOETVA hypocalcemia for the selected model.

Tables

(Tables are also embedded in the Results section above per journal convention; ensure each table fits within one page and is provided in editable format, not as an image)

1 **Table 1.** Preoperative characteristics by hypocalcemia status.

2 **Table 2.** Optimism-corrected internal validation performance of the three candidate models.

3 Compliance Checklist (for authors — remove before final submission)

- 4 • Title ≤ 20 words: 13 words — ✓
- 5 • Abstract ≤ 250 words, structured, no references — ✓ (verify word count once Results filled in)
- 6 • Keywords ≤ 6 — ✓ (6 keywords)
- 7 • Main text (Intro+Results+Discussion) $\leq 4,500$ words — check once complete
- 8 • References ≤ 60 , Vancouver style, numbered — ✓ (8 so far)
- 9 • Discussion has no subheadings — ✓
- 10 • Display items (figures+tables) ≤ 8 total — currently 2 tables + 2 figures = 4 — ✓
- 11 • No footnotes used — ✓
- 12 • Title page with affiliations and corresponding author asterisk — ✓ (fill in placeholders)
- 13 • Ethics declaration + consent statement included in Methods — ✓ (fill in placeholders)
- 14 • Author contributions, competing interests, funding statements included — ✓ (fill in placeholders)
- 15 • Line numbers included — ✓ (added at section level)
- 16 • Single manuscript file ≤ 10 MB combining text + figures for first submission — to confirm before final
- 17 packaging